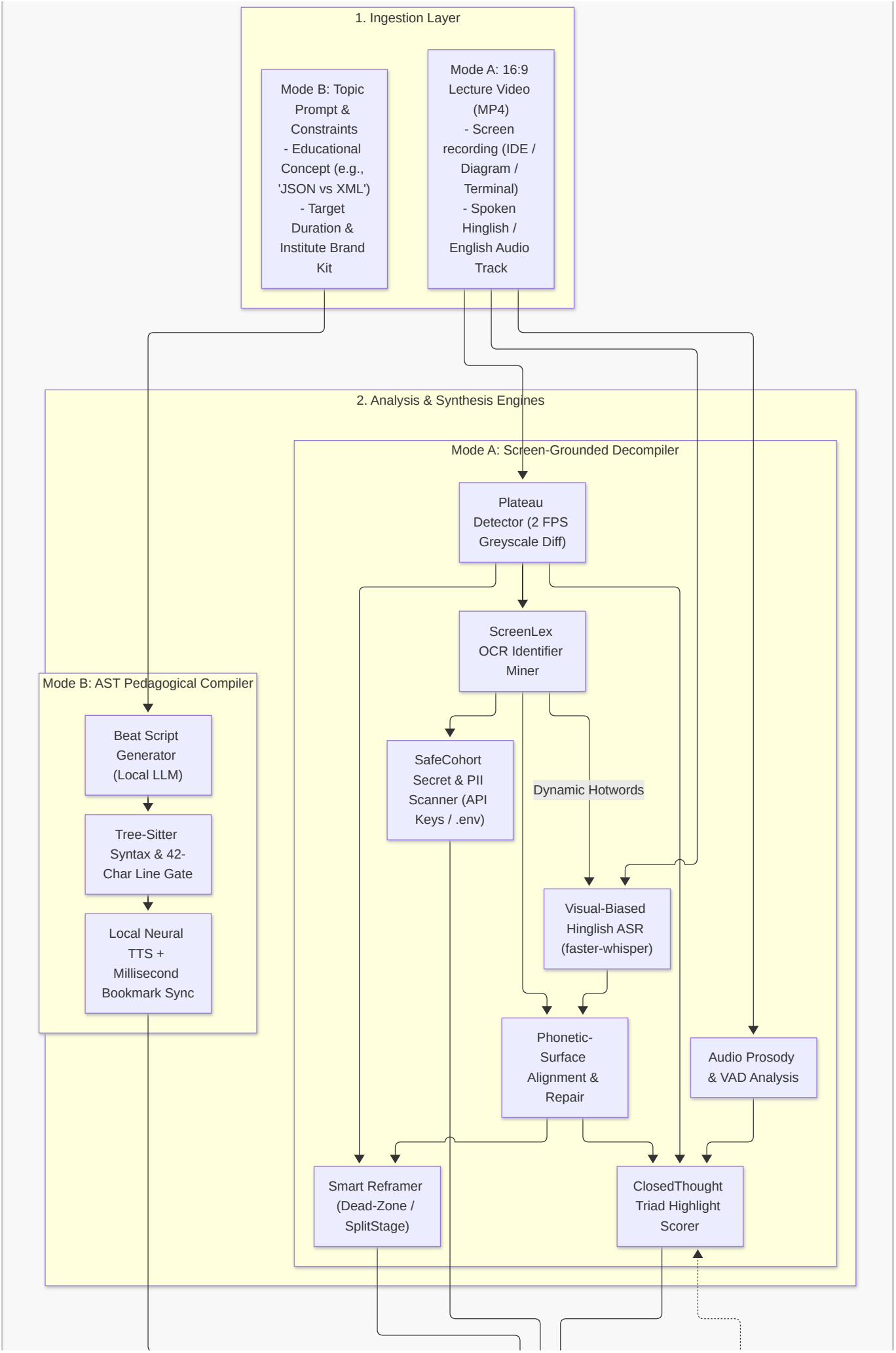
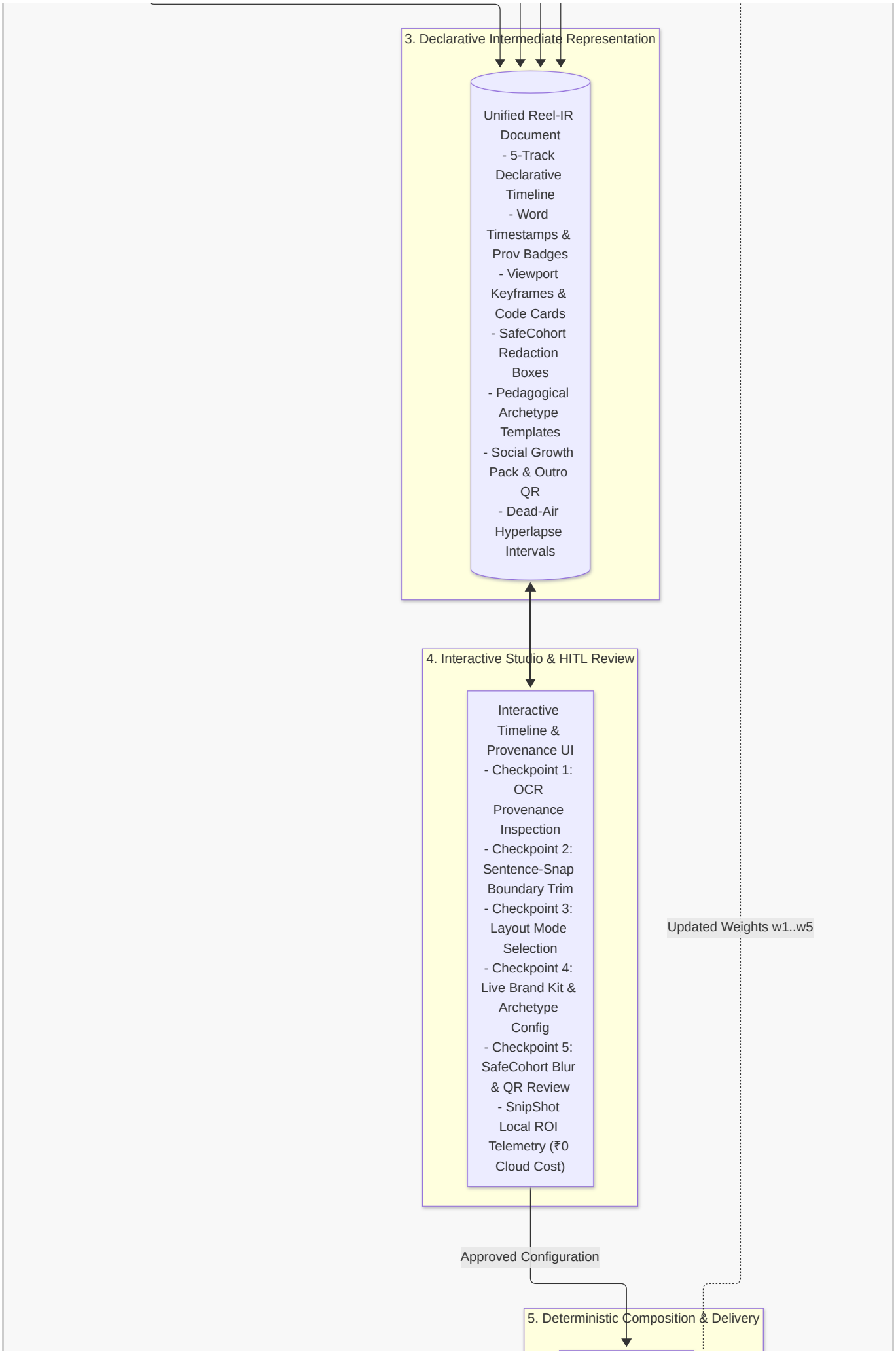
SnipShot (Reel Studio): Operational Workflows & System Pipelines

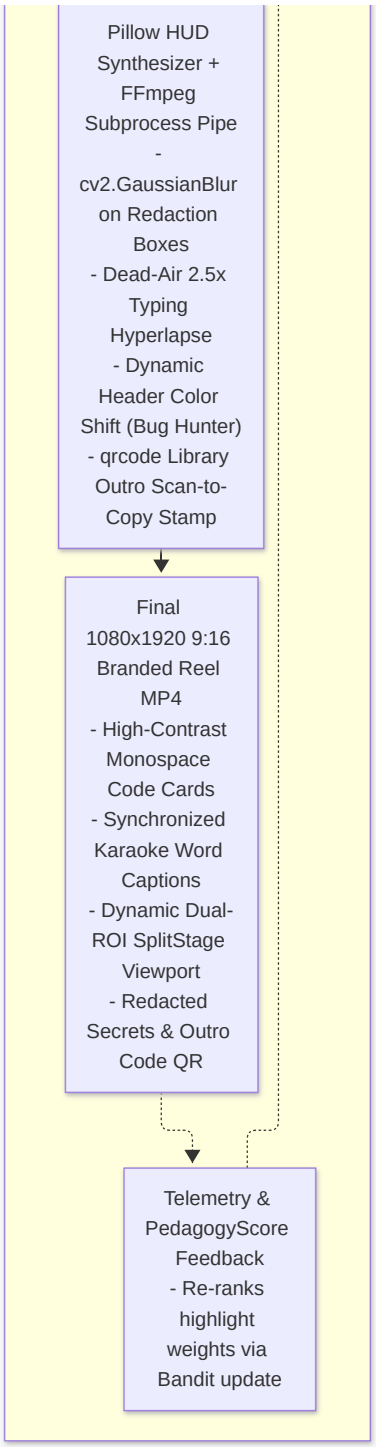
Document Version: 1.0.0
Status: Approved Operational Blueprint
Diagram Standard: Mermaid.js (Compatible with GitHub Markdown & Antigravity Studio)

1. Bird's-Eye Operational Workflow

This high-level flowchart illustrates the end-to-end lifecycle of media assets inside **SnipShot**, mapping the path from raw inputs (Mode A lecture recordings or Mode B topic prompts) through the processing engines, the unified `Reel-IR` bridge, the interactive Studio, and final video composition.


1. Ingestion Layer
Mode B: Topic Prompt & Constraints - Educational Concept (e.g., 'JSON vs XML') - Target Duration & Institute Brand Kit
Mode A: 16:9 Lecture Video (MP4) - Screen recording (IDE / Diagram / Terminal) - Spoken Hinglish / English Audio Track
2. Analysis & Synthesis Engines
Mode A: Screen-Grounded Decompiler
Plateau Detector (2 FPS Greyscale Diff)
ScreenLex OCR Identifier Miner
SafeCohort Secret & PII Scanner (API Keys / .env)
Dynamic Hotwords
Visual-Biased Hinglish ASR (faster-whisper)
Phonetic-Surface Alignment & Repair
Audio Prosody & VAD Analysis
Smart Reframer (Dead-Zone / SplitStage)
ClosedThought Triad Highlight Scorer
Mode B: AST Pedagogical Compiler
Beat Script Generator (Local LLM)
Tree-Sitter Syntax & 42-Char Line Gate
Local Neural TTS + Millisecond Bookmark Sync

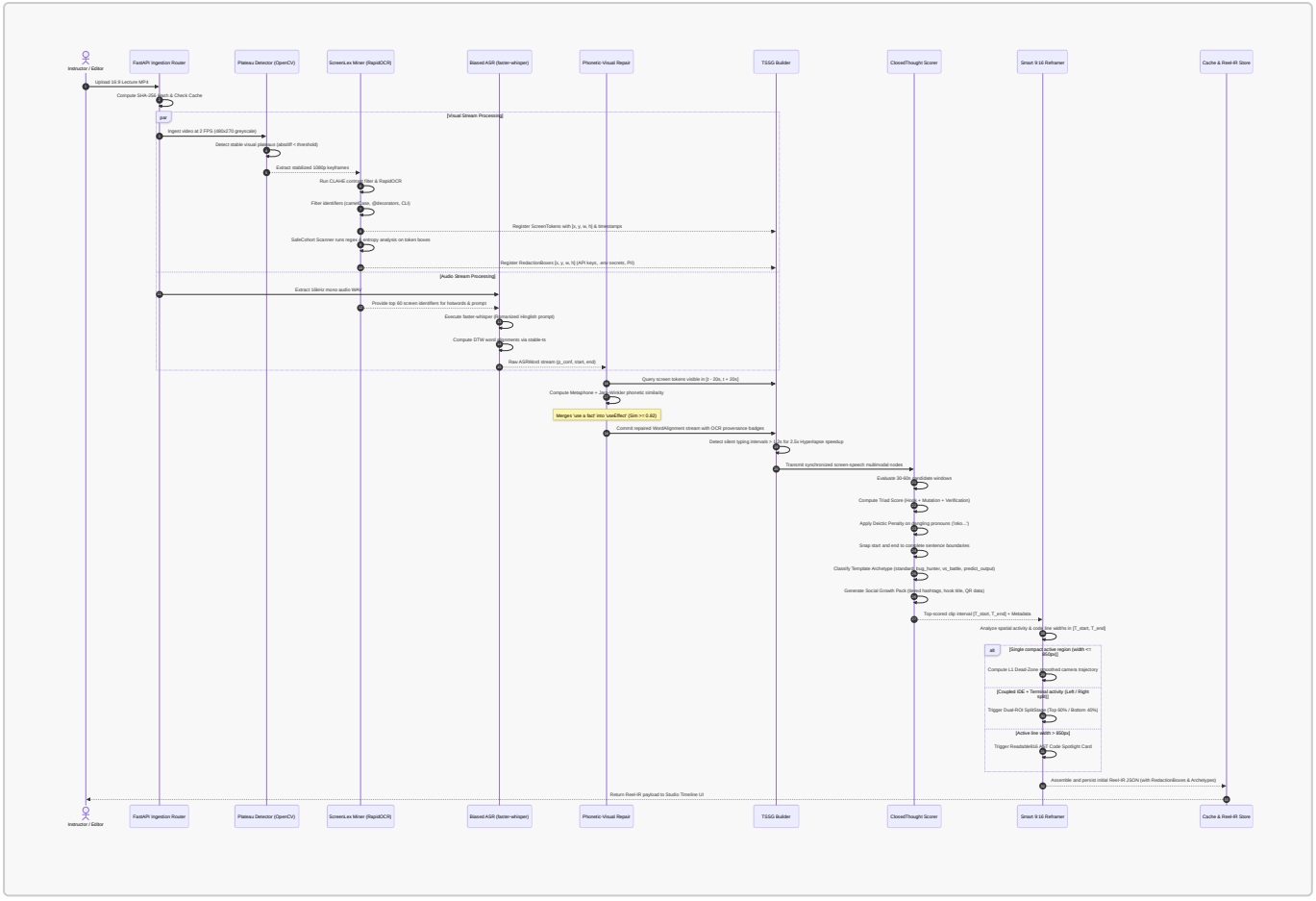

3. Declarative Intermediate Representation
Unified Reel-IR Document
- 5-Track Declarative Timeline
- Word Timestamps & Prov Badges
- Viewport Keyframes & Code Cards
- SafeCohort Redaction Boxes
- Pedagogical Archetype Templates
- Social Growth Pack & Outro QR
- Dead-Air Hyperlapse Intervals
4. Interactive Studio & HITL Review
Interactive Timeline & Provenance UI
- Checkpoint 1: OCR Provenance Inspection
- Checkpoint 2: Sentence-Snap Boundary Trim
- Checkpoint 3: Layout Mode Selection
- Checkpoint 4: Live Brand Kit & Archetype Config
- Checkpoint 5: SafeCohort Blur & QR Review
- SnipShot Local ROI Telemetry (₹0 Cloud Cost)
Updated Weights w1..w5
Approved Configuration
5. Deterministic Composition & Delivery



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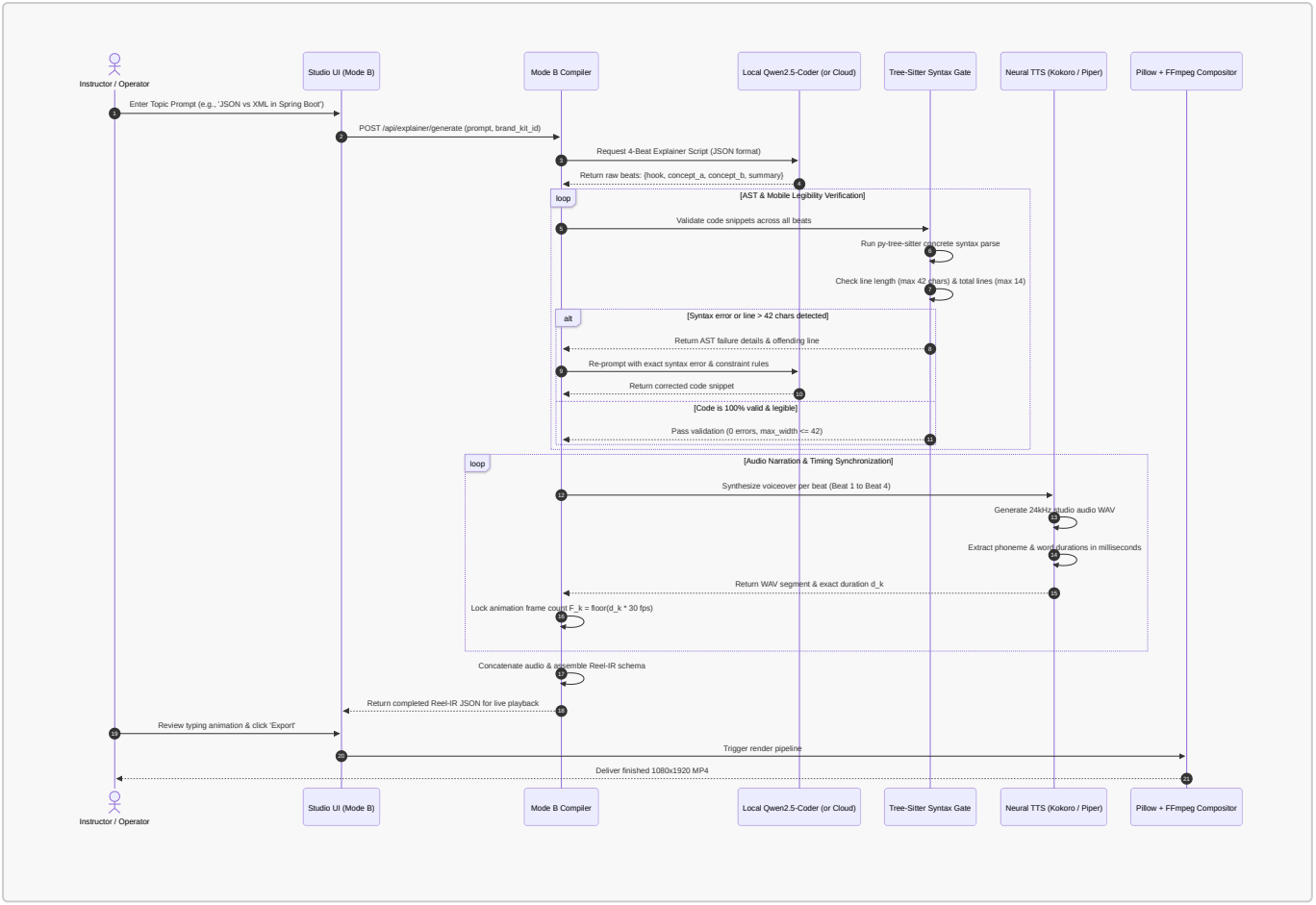
2. Mode A Pipeline: Screen-Grounded Decompiler

Mode A processes an unedited 16:9 lecture video into complete-thought 9:16 educational clips. The sequence diagram below shows how the visual screen stream and audio stream interact through the **Time-Aligned Screen-Speech Graph (TSSG)**.



3. Mode B Pipeline: AST-Verified Pedagogical Compiler

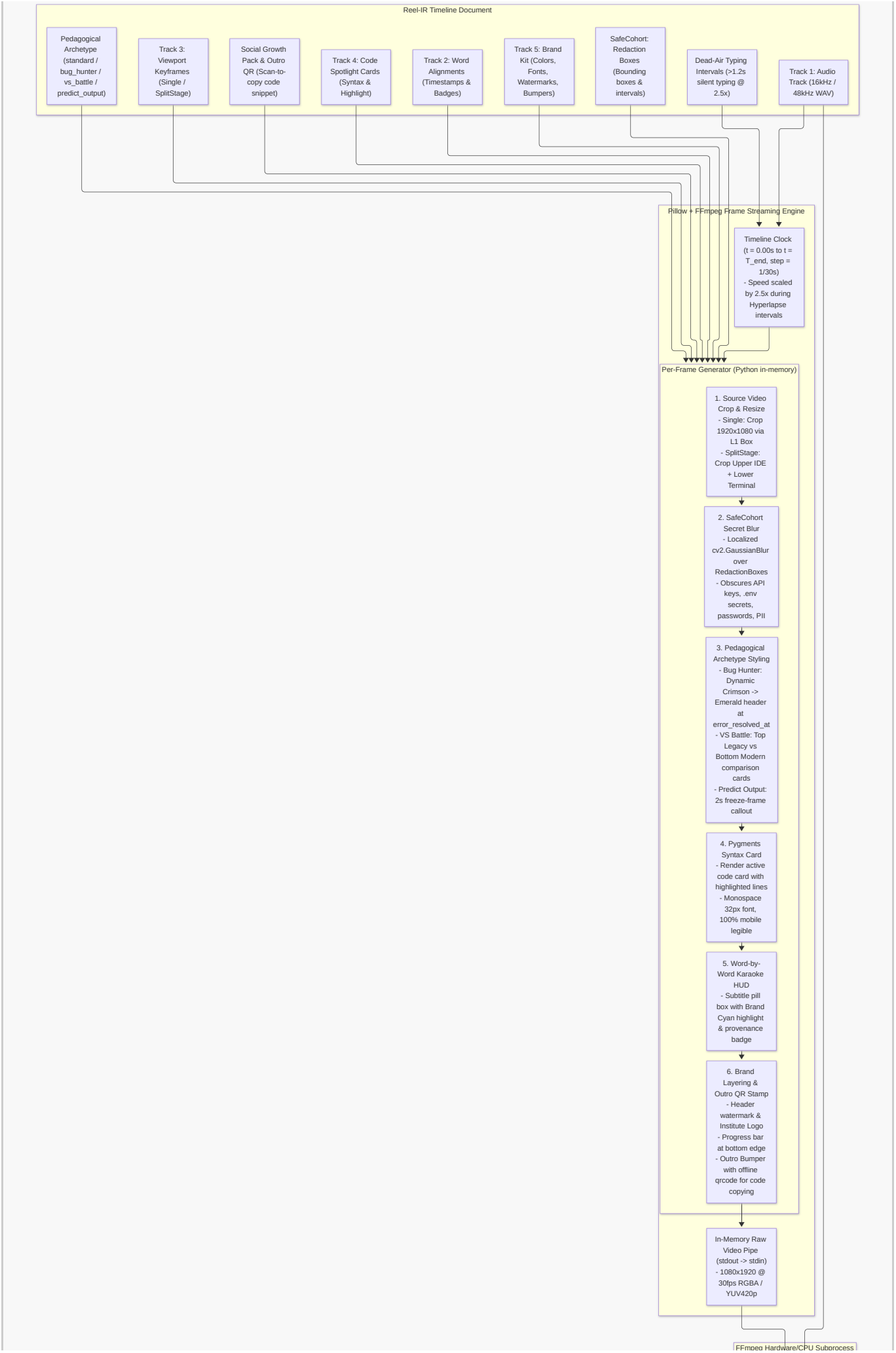
Mode B compiles an educational topic prompt into a high-trust, syntax-validated 9:16 code explainer reel. The pipeline enforces a strict AST gate to ensure that no broken code or illegible lines ever reach production.

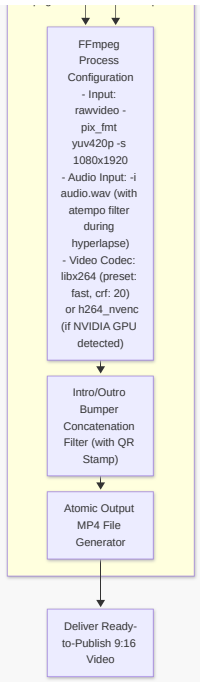


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4. Video Rendering & Composition Pipeline

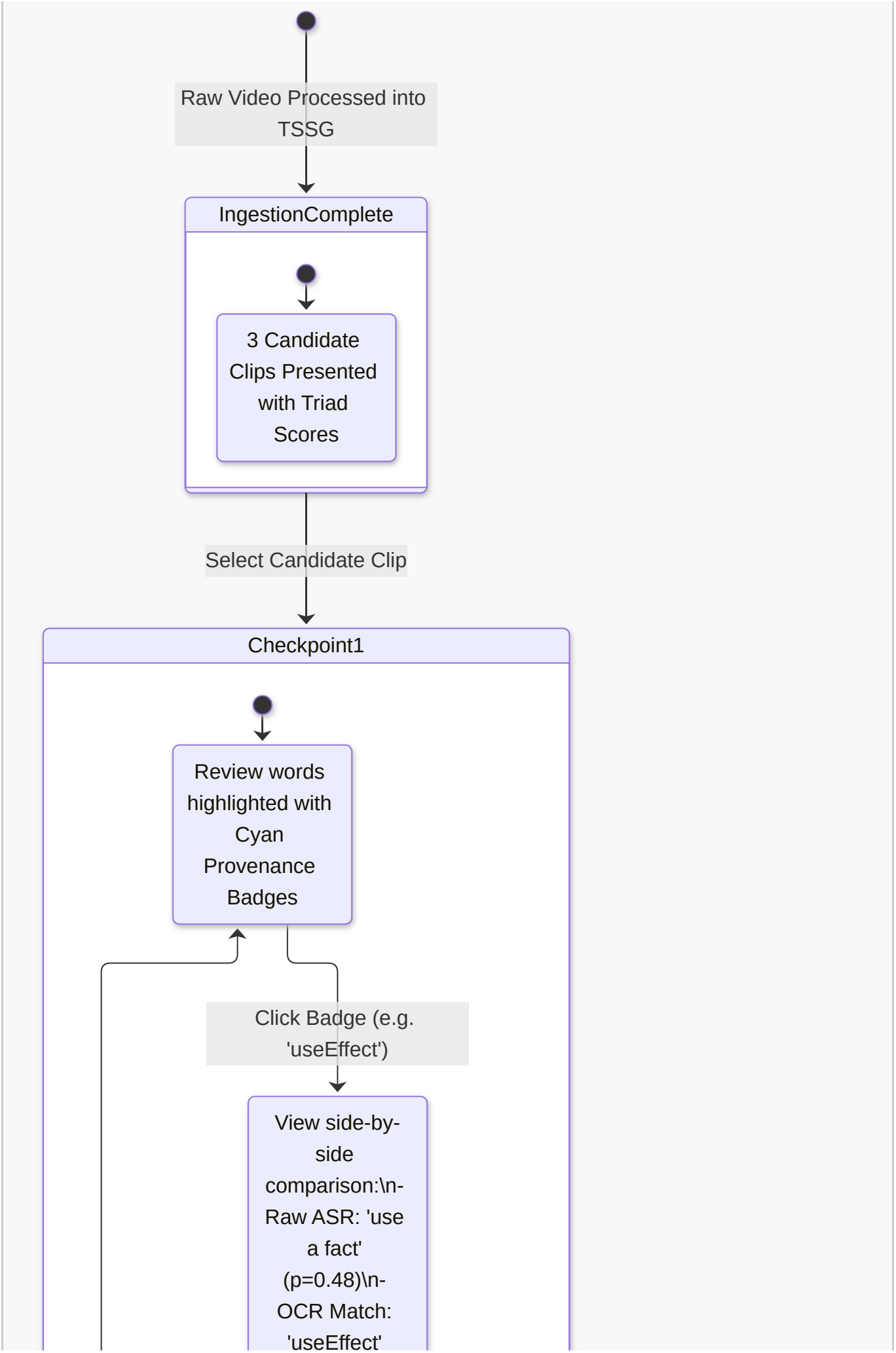
This flowchart details how the declarative 5-track `Reel-IR` document is transformed into a broadcast-quality 1080×1920 MP4 video without using heavy browser-based tools.

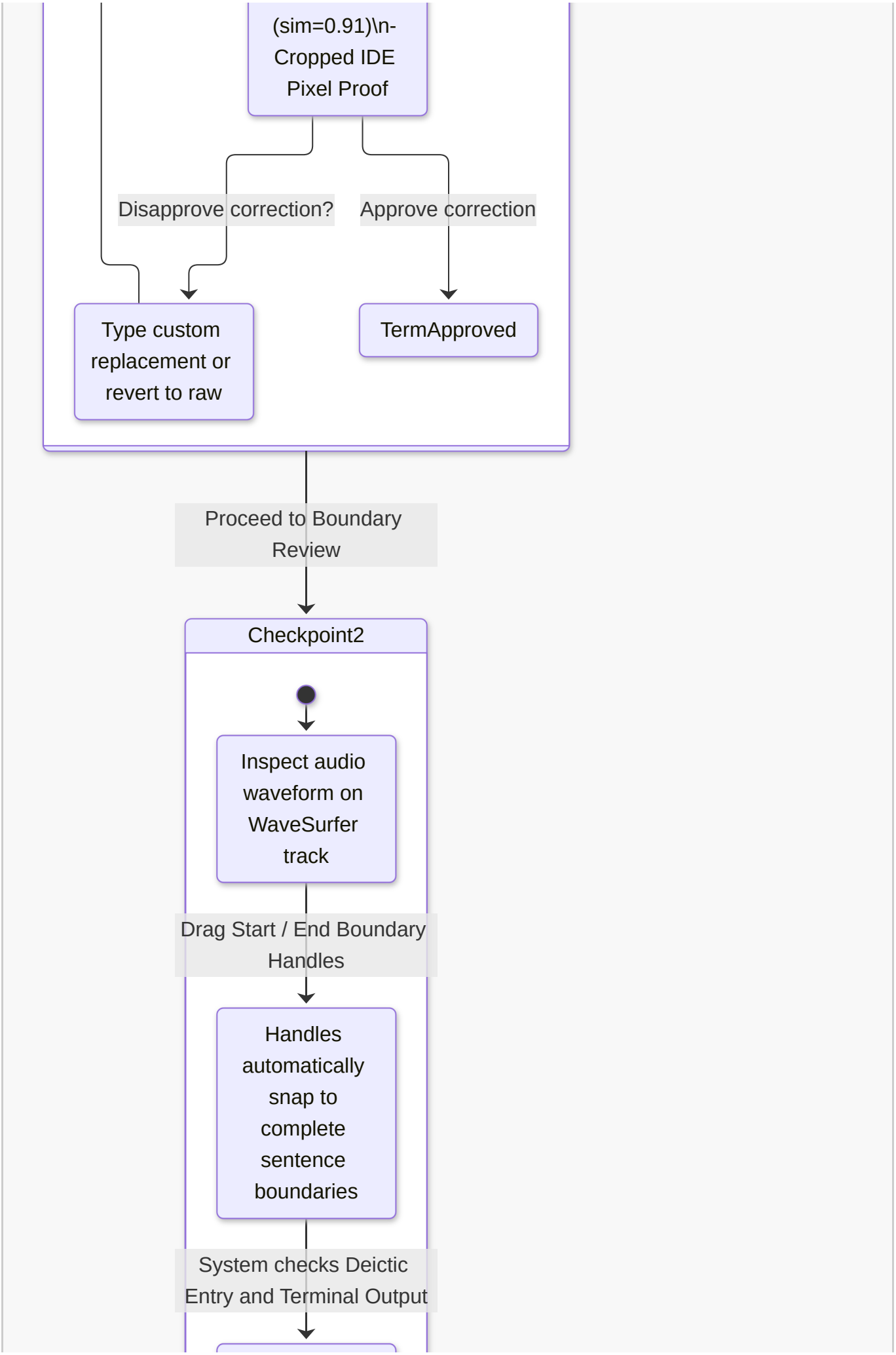

Reel-IR Timeline Document
Pedagogical Archetype (standard / bug_hunter / vs_battle / predict_output)
Track 3: Viewport Keyframes (Single / SplitStage)
Social Growth Pack & Outro QR (Scan-to-copy code snippet)
Track 4: Code Spotlight Cards (Syntax & Highlight)
Track 2: Word Alignments (Timestamps & Badges)
Track 5: Brand Kit (Colors, Fonts, Watermarks, Bumpers)
SafeCohort: Redaction Boxes (Bounding boxes & intervals)
Dead-Air Typing Intervals (>1.2s silent typing @ 2.5x)
Track 1: Audio Track (16kHz / 48kHz WAV)
Pillow + FFmpeg Frame Streaming Engine
Timeline Clock (t = 0.00s to t = T_end, step = 1/30s)
- Speed scaled by 2.5x during Hyperlapse intervals
Per-Frame Generator (Python in-memory)
1. Source Video Crop & Resize
- Single: Crop 1920x1080 via L1 Box
- SplitStage: Crop Upper IDE + Lower Terminal
2. SafeCohort Secret Blur
- Localized cv2.GaussianBlur over RedactionBoxes
- Obscures API keys, .env secrets, passwords, PII
3. Pedagogical Archetype Styling
- Bug Hunter: Dynamic Crimson -> Emerald header at error_resolved_at
- VS Battle: Top Legacy vs Bottom Modern comparison cards
- Predict Output: 2s freeze-frame callout
4. Pygments Syntax Card
- Render active code card with highlighted lines
- Monospace 32px font, 100% mobile legible
5. Word-by-Word Karaoke HUD
- Subtitle pill box with Brand Cyan highlight & provenance badge
6. Brand Layering & Outro QR Stamp
- Header watermark & Institute Logo
- Progress bar at bottom edge
- Outro Bumper with offline qrcode for code copying
In-Memory Raw Video Pipe (stdout -> stdin)
- 1080x1920 @ 30fps RGBA / YUV420p
FFmpeg Hardware/CPU Subprocess

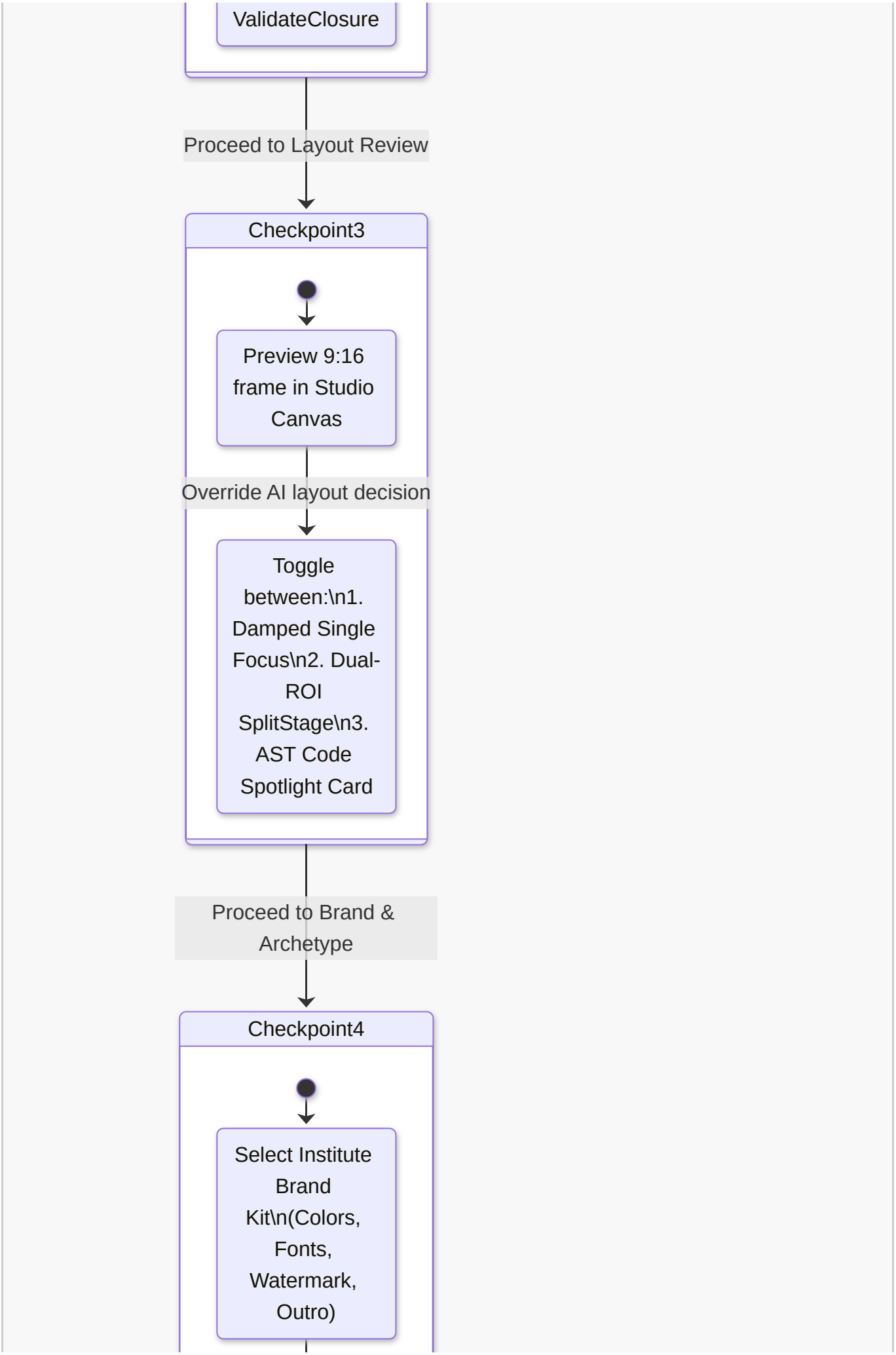


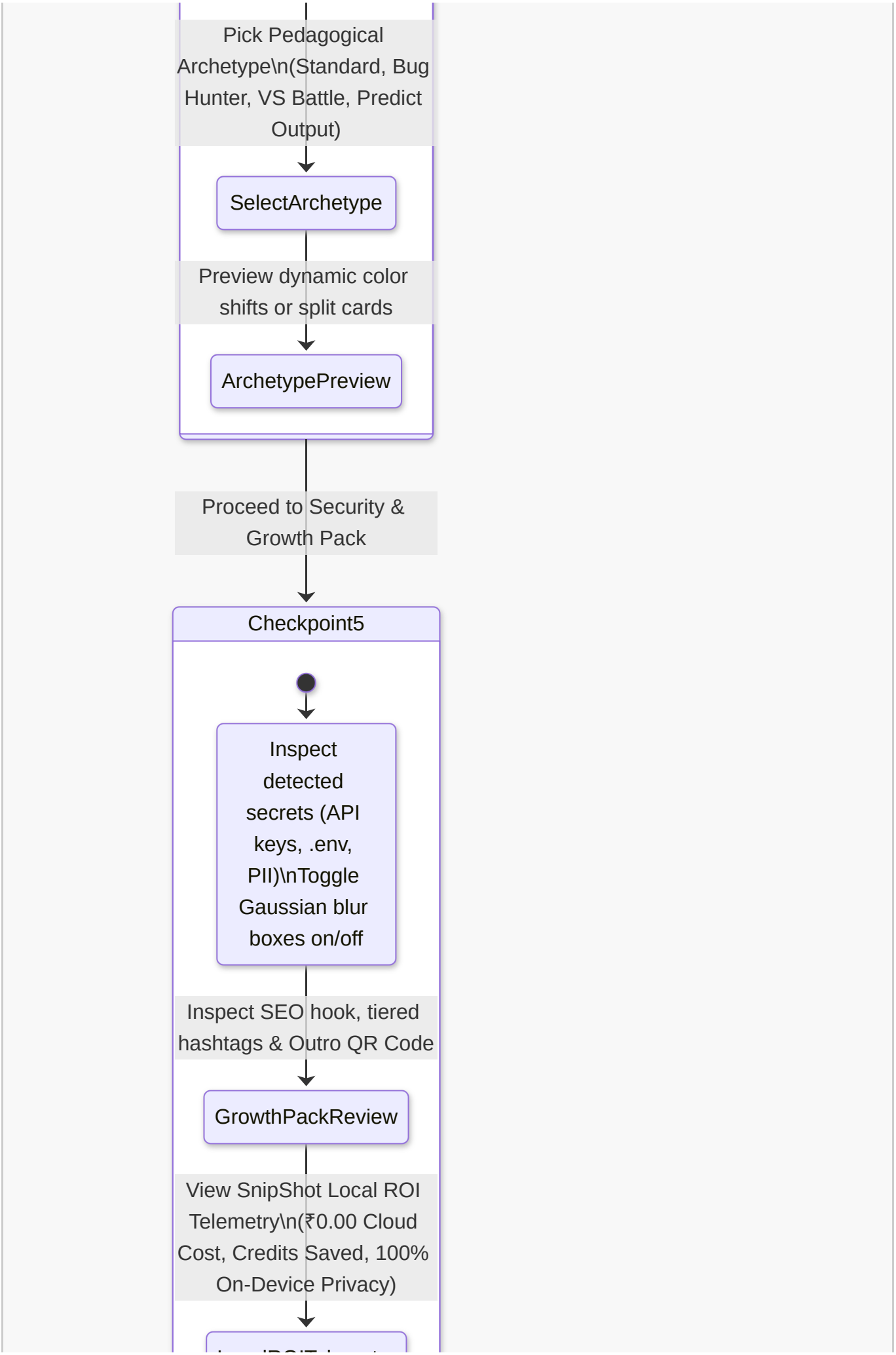
5. Human-in-the-Loop (HITL) Validation Checkpoints

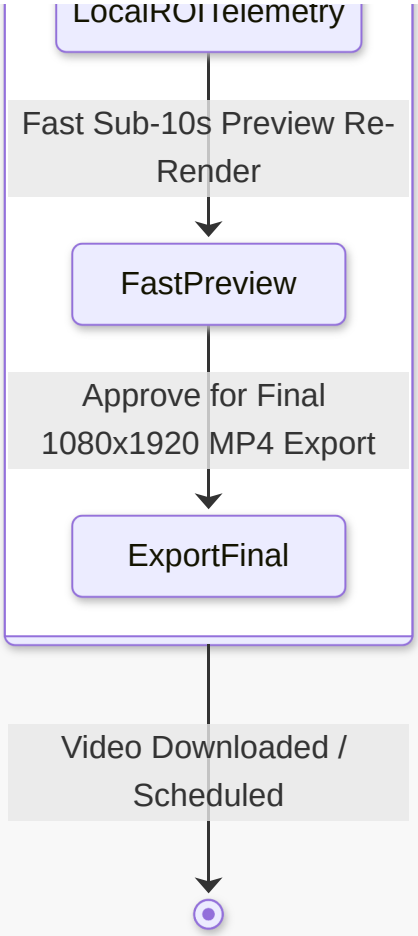
SnipShot integrates five explicit validation checkpoints where the instructor or editor can inspect, adjust, and approve AI-generated choices before video export.


Raw Video Processed into TSSG
IngestionComplete
3 Candidate Clips Presented with Triad Scores
Select Candidate Clip
Checkpoint1
Review words highlighted with Cyan Provenance Badges
Click Badge (e.g. 'useEffect')
View side-by-side comparison:\n- Raw ASR: 'use a fact' (p=0.48)\n- OCR Match: 'useEffect'


(sim=0.91)\n- Cropped IDE Pixel Proof
Disapprove correction?
Approve correction
Type custom replacement or revert to raw
TermApproved
Proceed to Boundary Review
Checkpoint2
Inspect audio waveform on WaveSurfer track
Drag Start / End Boundary Handles
Handles automatically snap to complete sentence boundaries
System checks Deictic Entry and Terminal Output

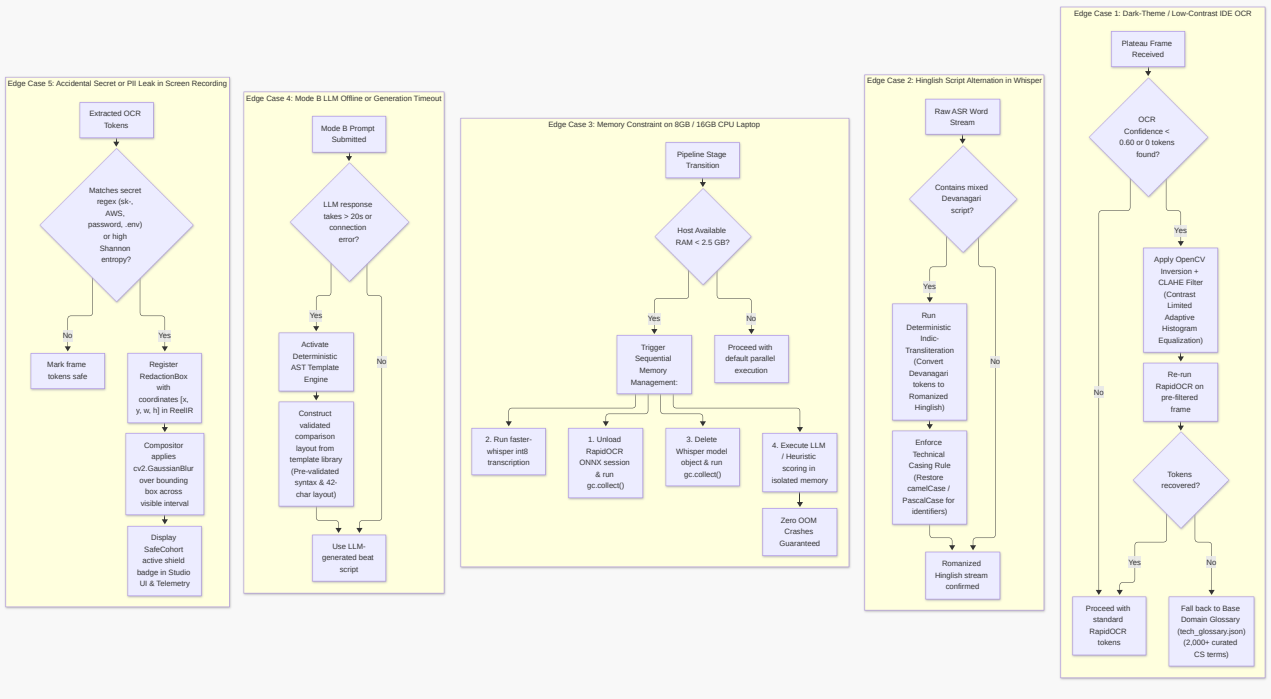

ValidateClosure
Proceed to Layout Review
Checkpoint3
Preview 9:16 frame in Studio Canvas
Override AI layout decision
Toggle between:\n1. Damped Single Focus\n2. Dual-ROI SplitStage\n3. AST Code Spotlight Card
Proceed to Brand & Archetype
Checkpoint4
Select Institute Brand Kit\n(Colors, Fonts, Watermark, Outro)


Pick Pedagogical Archetype\n(Standard, Bug Hunter, VS Battle, Predict Output)
SelectArchetype
Preview dynamic color shifts or split cards
ArchetypePreview
Proceed to Security & Growth Pack
Checkpoint5
Inspect detected secrets (API keys, .env, PII)\nToggle Gaussian blur boxes on/off
Inspect SEO hook, tiered hashtags & Outro QR Code
GrowthPackReview
View SnipShot Local ROI Telemetry\n(₹0.00 Cloud Cost, Credits Saved, 100% On-Device Privacy)



6. Error Handling & Fallback Decision Matrix

This flowchart details how the system gracefully handles real-world edge cases (dark themes, high CPU load, Hinglish script drift, LLM timeouts, and exposed secrets) without failing or producing corrupted outputs.



7. Telemetry & Continuous Optimization Feedback Loop

To continuously improve highlight selection accuracy, SnipShot includes a lightweight `PedagogyScore` feedback loop powered by a contextual bandit algorithm.

